Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

MATHEMATICS (Higher 2) 9758/02

Paper 2 **18 September 2026**

3 hours

Additional Materials : Printed Answer Booklet
List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Answer **all** questions.
Write your answers on the Printed Answer Booklet. Follow the instructions on the front cover of the answer booklet.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are **not** allowed in a question, you must present the mathematical steps using mathematical notations and not calculator commands.
You must show all necessary working clearly.

The number of marks is given in brackets [ ] at the end of each question or part question.

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Section A: Pure Mathematics [40 marks]

1 **(a)** Without using a calculator, solve the inequality $\dfrac{2x^2-3x+5}{x^2+3x-4}>1$. [4]

(b) Hence solve the inequality $\dfrac{2-3x+5x^2}{1+3x-4x^2}>1$. [3]

2 With reference to the origin O, a variable point R has position vector $\mathbf{r}$ and the fixed points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-zero and non-parallel vectors.

(a) Give a geometrical interpretation of $|\mathbf{a}\times\mathbf{b}|$. [1]

(b) Given that $3\mathbf{r}\times(\mathbf{b}-\mathbf{a})=\mathbf{0}$, describe geometrically the set of all possible positions of the point R. [2]

(c) Given that $\mathbf{r}\cdot\mathbf{b}=\mathbf{r}\cdot\mathbf{a}$, describe geometrically the set of all possible positions of the point R. [2]

The point C has position vector $\begin{pmatrix}-3\\-6\\2\end{pmatrix}$.

(d) Find the sine of the acute angle between $\overrightarrow{OC}$ and the x-axis, leaving your answer in the form of $k\sqrt{10}$, where k is an exact value to be determined. [3]

3 A curve C has equation $(x+y)^2+axy=6$, where a is a positive constant. A line L has equation $y=-x+k$, intersects curve C for all $|k|\geq 2$ where $k\in\mathbb{R}$.

(a) Show that the x-coordinate of the point(s) of intersection would satisfy the equation

$$ax^2-akx+(6-k^2)=0.$$ [1]

(b) Find the value of a. [4]

4 A curve C is defined by the parametric equations

$$x = \sin t + \sqrt{3}\cos t, \quad y = \sqrt{3}\cos t, \quad 0 \le t \le \pi.$$

(a) Express x in the form of $R\sin(t+\alpha)$, where R is an exact positive constant and α is an acute angle to be found. [2]

(b) Deduce the exact maximum value of x on C and explain why the minimum value of x on C is not $-R$. [2]

(c) Sketch the graph of C, showing the exact coordinates of the endpoints and the points of intersection with the x and y axes. [3]

(d) Find the cartesian equation of C in the form of $x = \mathrm{f}(y)$. State clearly the restriction on the values of y. [3]

5 At a high-tech urban farm, a recycling tank initially contains 200 litres of pure water. To support crop growth, a nutrient solution is pumped into the tank at a constant rate of 10 litres per minute. The concentration of nitrogen in the incoming solution is 0.075 grams per litre. The well-mixed solution is pumped out of the tank at the same rate of 10 litres per minute. Let x be the mass (in grams) of nitrogen in the tank at time t minutes.

(a) Show that the rate of change of the mass of nitrogen is given by the differential equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{3}{4} - \frac{1}{20}x.$$ [2]

(b) It is given that the tank initially contains pure water. Solve the differential equation in part **(a)** to find an expression for x in terms of t. [5]

(c) Sketch the graph of x against t. Explain in this context what happens to the nitrogen mass in the long run. [3]

Section B: Probability and Statistics [60 marks]

6 Tom has a box containing 90 ornaments that are indistinguishable aside from their colours and pictorial designs on them. The numbers of each colour and pictorial design are given in the table.

	White	Blue	Pink	Green
Star	12	6	9	13
Diamond	2	3	7	10
Moon	4	5	11	8

Tom chooses two items from his box of ornaments at random.

(a) Given that neither of the items chosen are green, find the probability that both the items chosen are either white or has a moon pictorial design (or both). [2]

(b) Find the probability that the two items include exactly one blue ornament and exactly one ornament that has a star pictorial design. [3]

(c) Tom has a favourite type of ornament out of the 12 possible colour / pictorial design combinations and the chance of him getting both items being his favourite type of ornament is $\frac{28}{4005}$. Determine Tom's favourite type of ornament, showing your working clearly. [2]

7 Ten cards each bear a single letter, and together they can be made to spell the word "ASSESSMENT". Cards bearing the same letter are not distinguishable.

(a) Four cards are selected from the ten cards, and the order of selection is not relevant. Find the number of possible selections without restriction. [3]

(b) All ten cards are arranged in a row.

(i) Find the number of arrangements in which the letters "A", "M", "N" and "T" are together. [2]

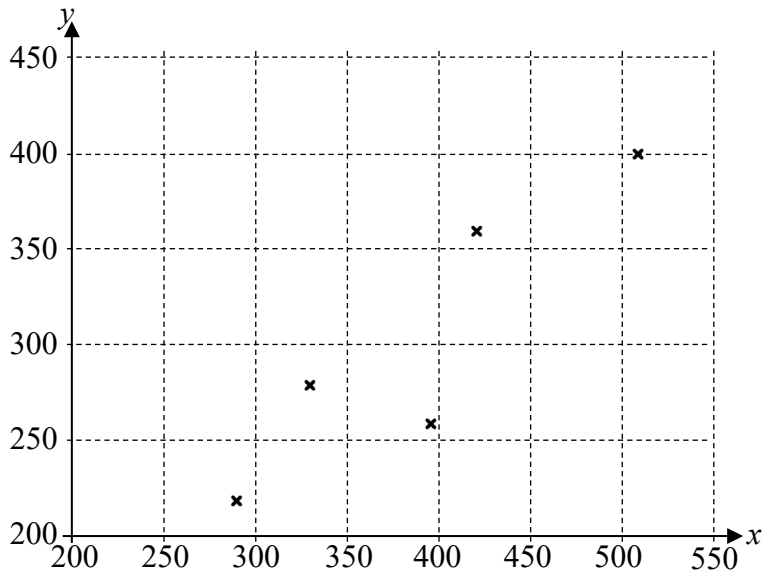
(ii) Find the number of arrangements in which no two letters "S" are together. [2]

(iii) Hence find the number of arrangements in which no two adjacent letters are the same. [2]

8 Zayne compares the brightness readings given by two different devices. At five different locations with different lighting conditions, he measured the brightness using the two different devices. The table shows the brightness reading from a laboratory light meter is recorded as x units and the brightness reading from a phone light sensor app is recorded as y units.

x	289	329	395	420	508
y	220	280	260	360	400

A scatter diagram for the data is shown below.



The school wants to use the more convenient phone app reading to estimate what the laboratory light meter reading would be.

(a) Zayne decides to investigate whether the model $x = 98 + 0.95y$ is a good fit for the data.

(i) In the Answer Booklet, draw the line $x = 98 + 0.95y$ on the scatter diagram given. [1]

(ii) To check the fit of the model, Zayne wants to analyse the residual for each point on the scatter diagram in part **(a)(i)**. For the model $x = \mathrm{f}(y)$, the residual for a point (a, b) is $a - \mathrm{f}(b)$. Explain why this definition of residual is appropriate in this context, and why Zayne should consider the sum of the squares of these residuals. [2]

(iii) Mark the residuals for each point on the scatter diagram in part **(a)(i)**. [1]

(iv) Calculate the sum of the squares of the residuals for the line $x = 98 + 0.95y$. [1]

(b) **(i)** Use your calculator to find the equation of the least squares regression line of x on y and state the product moment correlation coefficient. [2]

(ii) Calculate the sum of the squares of the residuals for this regression line and verify that it is less than the value found in part **(a)(iv)**. [1]

(iii) The equation of the regression line is used to estimate the brightness reading from a laboratory light meter when the reading from a phone light sensor app is 300 units. Explain whether you would expect this estimate to be reliable. [1]

9 A cheese company produces cheese blocks of two different sizes, Standard and Large. The masses, in kilograms (kg), of the Large cheese blocks follow the distribution $N(4.25, 0.15^2)$ and the masses, in kilograms (kg), of the Standard cheese blocks follow the distribution $N(2.85, 0.2^2)$.

(a) A restaurant purchases 90 of the Standard cheese blocks. Calculate the expected number of these blocks that have a mass greater than 2.9 kg. [2]

(b) Find the probability that the total mass of 10 randomly chosen Large cheese blocks differs by less than 2.3 kg from the total mass of 15 randomly chosen Standard cheese blocks. [3]

(c) State, in the context of this question, an assumption needed for your calculation in part **(b)**. [1]

The company finds that there is a demand for Small cheese blocks. These Small cheese blocks are produced by cutting the Large cheese blocks into three exactly equal masses, or the Standard cheese blocks into two exactly equal masses.

(d) Find the probability that the mass of a randomly chosen Small cheese block made from a Large cheese block is greater than the mass of one made from a Standard cheese block. You should ignore any wastage caused by cutting the cheese block. [4]

(e) Without doing any detailed calculation, explain how your answer in part **(d)** would change if each cut of a cheese block caused a small amount of wastage. [1]

10 A food delivery company claims that the average time taken to prepare an order is less than 12 minutes. The preparation time, x minutes, was recorded for a random sample of 50 orders, and the results were summarised as follows:

$$\sum(x-12) = -11.9, \quad \sum(x-12)^2 = 17.36.$$

(a) Explain the meaning of a random sample in the context of the question. [1]

(b) Find the unbiased estimates of the population mean and variance of the preparation time. [2]

(c) Carry out a hypothesis test on the company's claim. Find the p-value for this test and explain what it indicates about the company's claim. You should state your hypotheses and define any symbols you use. [4]

After improvements to the data collection process, another random sample of 80 orders was taken. The sample mean was k minutes and the sample standard deviation was 2.3 minutes. The company's claim was not supported at the 5% significance level.

(d) Find the range of values of k, giving your answer correct to 2 decimal places. [3]

(e) It is known that the unbiased estimate of population variance found based on the sample data is close to the population variance. Explain why the same test would not be valid if the data collection consists of 12 orders instead. State an assumption needed for the test based on this number of orders. [2]

11 A company manufactures identical marbles in three colours: blue, red and green.

(a) A bag contains one blue, one red and one green marble. The marbles are drawn one by one at random with replacement. The random variable X is the number of draws needed for any marble of the same colour to be drawn a second time. The two draws of the same-coloured marbles do not need to be consecutive.

(i) Show that $\mathrm{P}(X=4)=\frac{2}{9}$ and find the probability distribution of X. [2]

(ii) Find the exact values of $\mathrm{E}(X)$ and $\mathrm{Var}(X)$. [2]

The marbles are sold in boxes of 25. On average, 40% of the marbles are blue while the rest are red or green (non-blue). Assume that the number of blue marbles in a box has a binomial distribution.

(b) Find the probability that a randomly chosen box contains fewer blue marbles than non-blue marbles. [1]

(c) John buys 5 boxes of marbles. Find the probability that at least 3 boxes contain fewer blue marbles than non-blue marbles. [2]

(d) Find the probability that five randomly chosen boxes contain no more than 50 blue marbles in total. [2]

(e) A random sample of 30 boxes is chosen. Using an approximate distribution, find the probability that the mean number of blue marbles in one box is between 7 and 11. [3]

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